

Title of the Research Project: Plastic Pollution in The Sediment of Kaptai Lake

Background of the Study:

Plastic pollution is one of the most significant environmental issues affecting aquatic ecosystems worldwide (Cole et al., 2011). Plastics and microplastics accumulate in sediments, altering their physical and chemical properties, impacting benthic organisms, and entering the food chain (Thompson et 2004). Kaptai Lake, a vital freshwater resource for the local community, has been increasingly impacted by human activities, including tourism, urban runoff, and improper waste disposal. While research on plastic pollution has been conducted in various water bodies, limited attention has been given to its presence in sediment, especially in the context of lakes in developing regions.

Aims and Objectives:

The primary aim of this research is to evaluate the extent of plastic pollution in the sediment of Kaptai Lake. The objectives include:

1. Identifying and quantifying plastic debris and microplastics in the lake's sediment.
2. Analyzing the types, sources, and distribution patterns of plastic pollution.
3. Assessing the potential ecological risks posed by plastic pollution to aquatic organisms and the broader lake ecosystem.
4. Recommending strategies to mitigate plastic pollution in Kaptai Lake and similar freshwater ecosystems.

Methodology:

The study will employ a multi-step methodological framework:

1. Sampling: Sediment samples will be collected from multiple locations within Kaptai Lake using a systematic grid approach to ensure spatial representation.
2. Laboratory Analysis: Samples will undergo density separation, visual identification, and Fourier Transform Infrared Spectroscopy (FTIR) to detect and characterize plastics and microplastics.
3. Source Analysis: Potential sources of plastic pollution will be assessed through field surveys and by analyzing land-use patterns and human activities around the lake.
4. Risk Assessment: The ecological impact of the identified plastics will be evaluated based on their abundance, composition, and toxicological data from literature.

Expected Outcome and Contribution of the Research:

The study is expected to provide the first comprehensive dataset on plastic pollution in the sediment of Kapta Lake. The results will highlight the scale and sources of plastic contamination, contributing to a better understanding of the ecological implications for freshwater systems. By offering evidence-based recommendations for pollution control and management, the research will support local authorities, policymakers, and environmental organizations in designing sustainable strategies to mitigate plastic pollution. Furthermore, this research will contribute to the global knowledge pool on freshwater plastic pollution, with potential applicability to other lake ecosystems.

The Frame of the Research Work:

Steps	Month								
	1	2	3	4	5	6	7	8	9
Preparation and Sampling Design									
Fieldwork									
Laboratory Analysis									
Data Analysis and Interpretation									
Report Writing and Dissemination									

References:

Cole, M., et al. (2011). "Microplastics as contaminants in the marine environment: A review." Marine Pollution Bulletin, 62(12), 2588-2597.

Thompson, R.C., et al. (2004). "Lost at sea: Where is all the plastic?" Science, 304(5672), 838-838.

This research will bridge a critical knowledge gap in understanding sedimentary plastic pollution in Kapta Lake and will have a lasting impact on regional and global freshwater conservation efforts.

